

ON THE IMPACT

Describing the impact of my research to general public is sometimes challenging. I often describe it in the context of the threat of the climate change for food security. However, it is unlikely that even in the face of the worst crop failure, we will run out of food in the Netherlands.

Sorghum is a crucial crop in Sub-Saharan Africa, mainly grown by small-holder farmers. Witchweed infestation often ruins sorghum yields entirely. My work on witchweed-resistant sorghum can improve lives of many farmers.



Witchweed infestation disproportionately affects women. In many regions, it is primarily managed through hand weeding, a labor-intensive task traditionally performed by young women, often limiting their access to education. The development of alternative control strategies could, over time, reduce this burden and enhance educational opportunities for girls.



Sorghum growing in the fields in Zimbabwe (photo by Thierfelder/CIMMYT)

ON PREVIOUS EXPERIENCES

I have taught multiple courses to Biology students in my career. I think that teaching makes me a better researcher and the other way around.



Creative solutions help convey the message. I once gave a research talk under the name *Striga hermonthica* (latin name of witchweed) not as Dorota Kawa.

Dr. Dorota Kawa

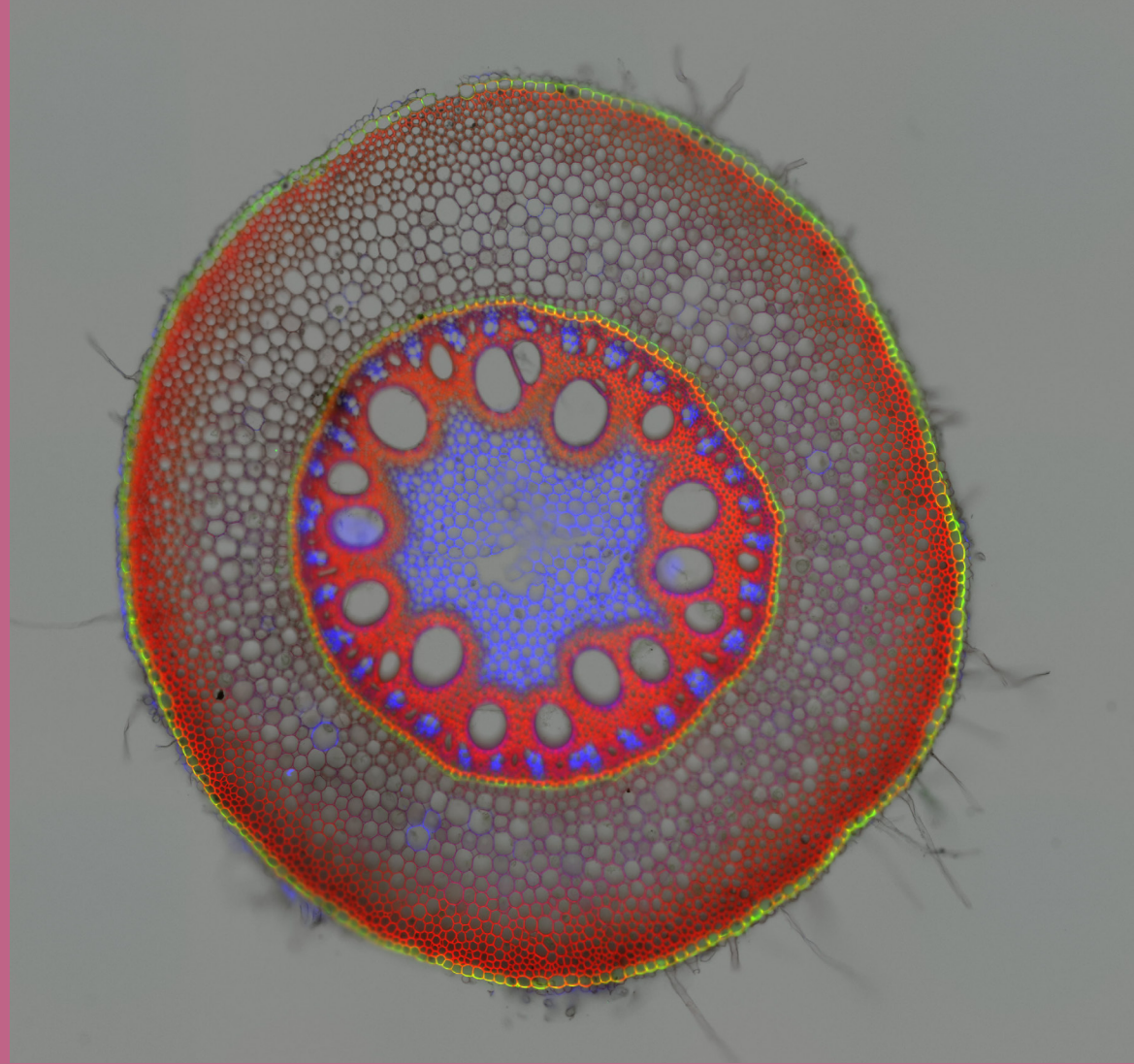
In my research, I study how plants like sorghum and its relative maize decide to make and wear protective layers made of suberin (cork) in their roots. These layers can protect a plant from drought, salinity or witchweed infection. We found some bacteria, naturally present in the soil, that can dress up plant roots with these stress-protective layers! I am also part of the BarrierFates consortium.



Can plant research have an impact on gender imbalances?

ON BARRIERFATES

BarrierFates takes our fundamental questions and applies them to answer how we could have a more stable food supply through more resilient crops.



Maize root cross section stained with different stains seen under the microscope. This way we can identify different protection mechanisms by the plant.

ON THE FUTURE

I think there are a few things we can do as scientists for science communication:

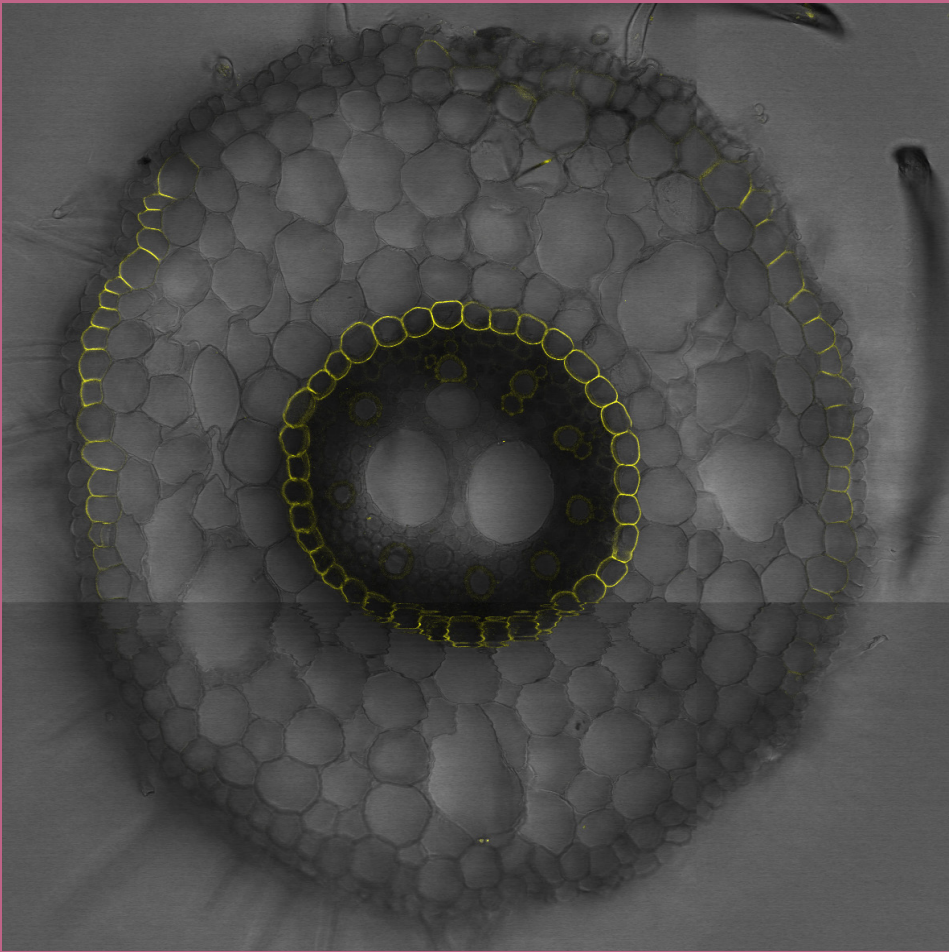
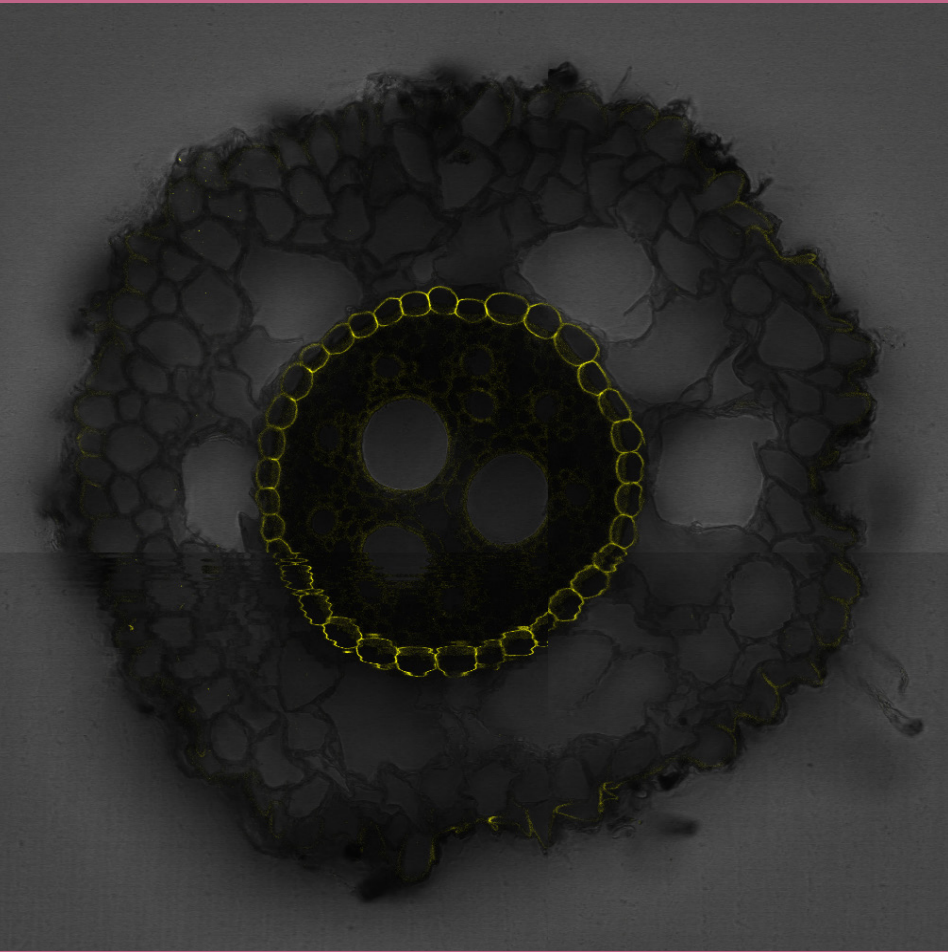
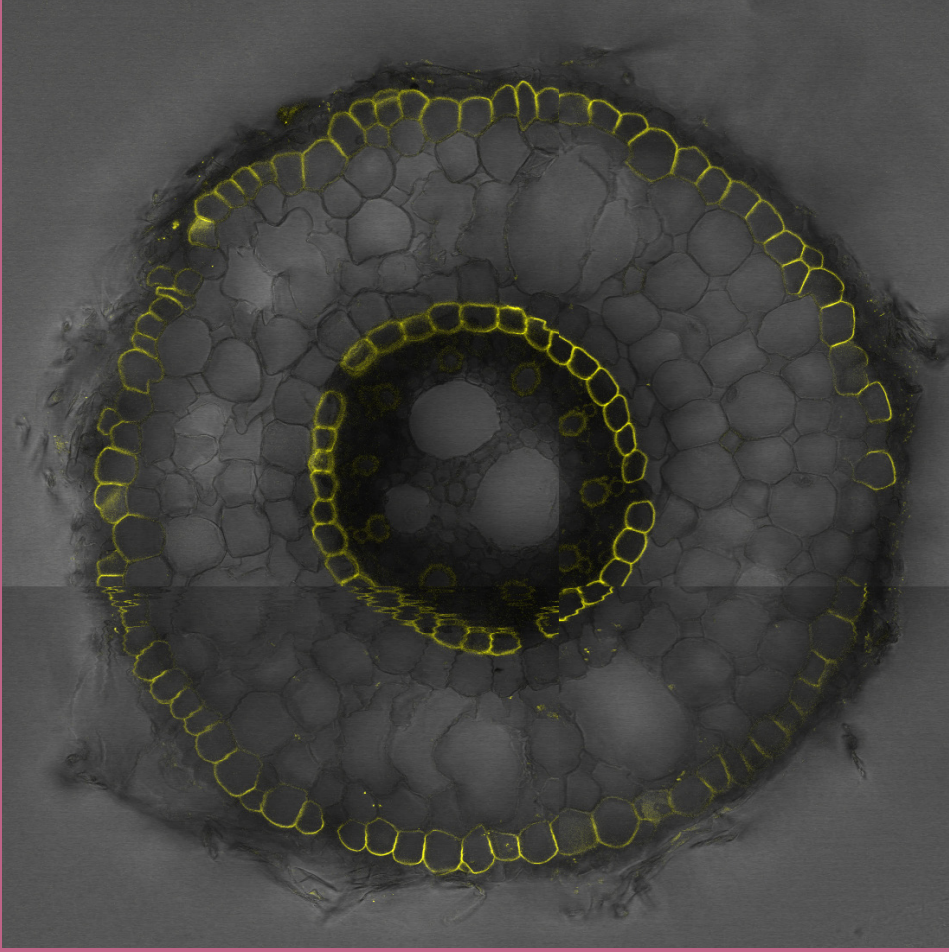
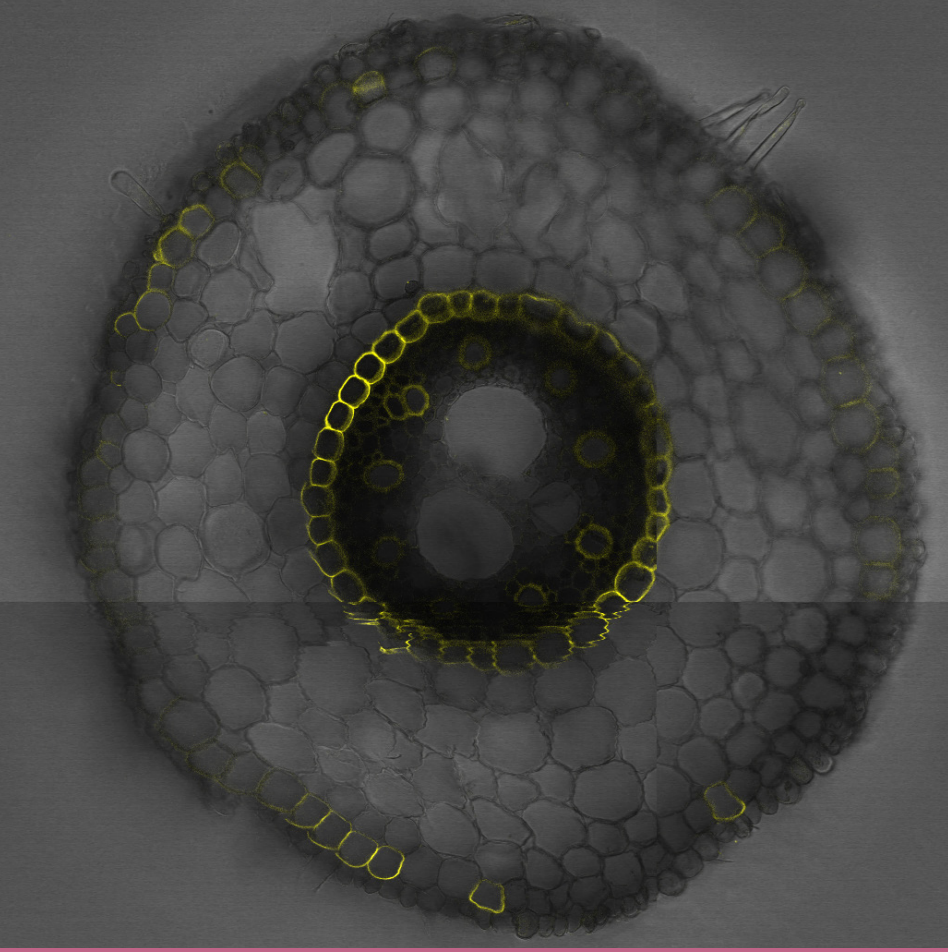
Show how beautiful roots can be. I hope that through this beauty the general public can see our scientific passion and appreciate fundamental aspects of our studies as well.

Work with social scientists, since researchers alone lack the expertise to address societal impact.

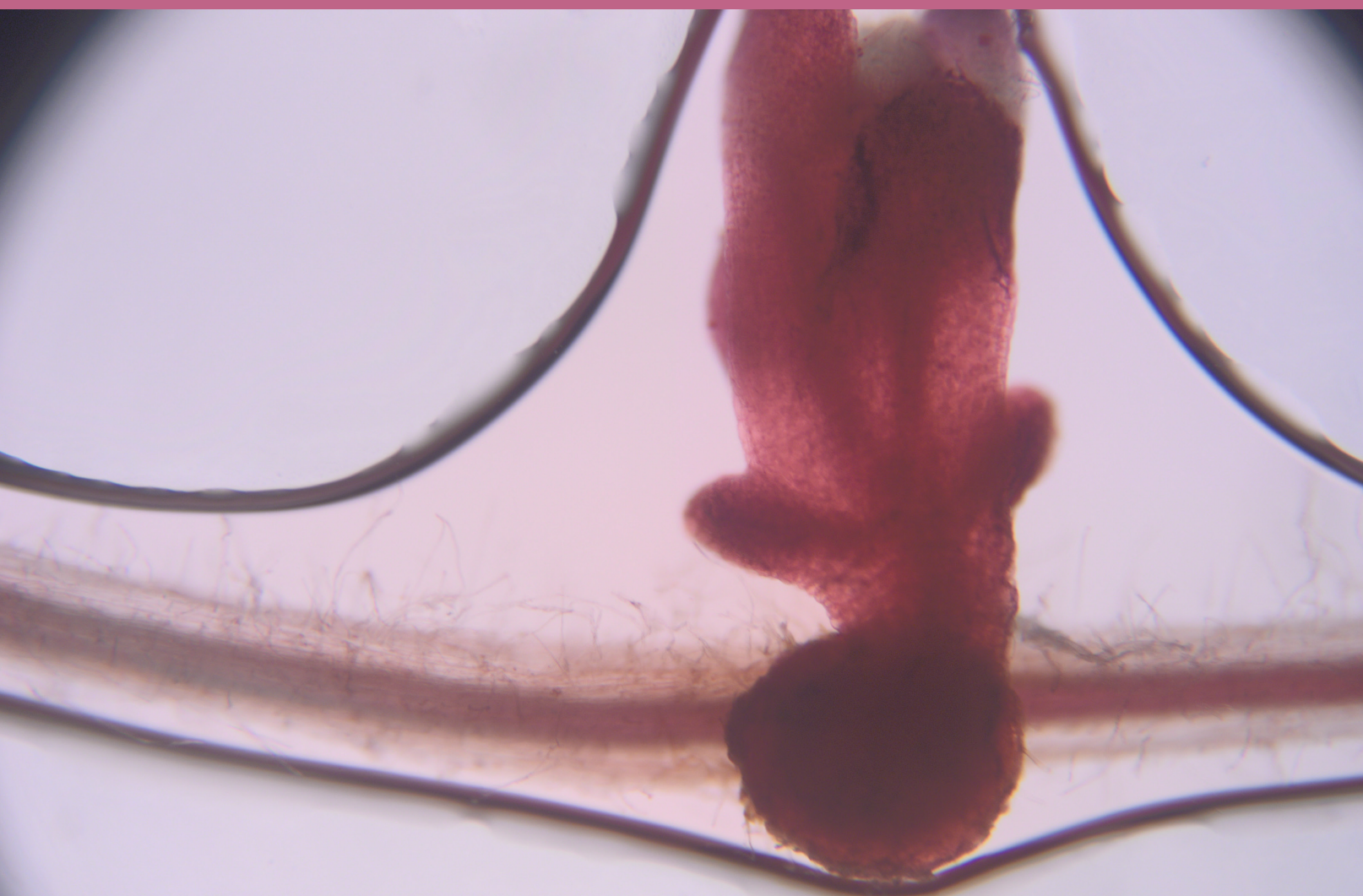
ON THE CHALLENGES

Outreach is a difficult task we’re not specifically trained for. It’s hard to be a researcher, teacher and science communicator all at one, and do it all well. While we often say our goal is to improve drought tolerance, our actual work focuses on understanding specific mechanisms in plants. That has clear scientific value, but only a long-term societal impact.

Many researchers often give similar answers: better yields, protecting biodiversity, improving crops, etc., because we’re trained to frame our work in this way. In reality, the most immediate value is often scientific and personal development of the researcher. Any long-term societal impact depends on the combined work of many researchers team members from various labs.



Microscope images of sorghum cross sections stained to identify cork (by Julia Mars). The yellow areas are where cork deposits to protect the roots from environmental stress.



Two plants connected. Witchweed steals nutrients from sorghum by connecting to the roots of this crop. This is a microscope image of this situation. The horizontal root is of sorghum, while the other figure in the image is witchweed.

Read this poster in Dutch



What do you think?

